

Input data and their popularity

Q: What is the founded by Welch's?

A: Thomas Bramwell Welch
popularity score (s_i):

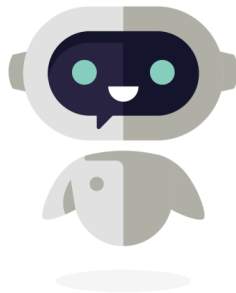
44

Q: What is the capital of Poland?

A: Warsaw
popularity score (s_i):

2443

AdaPop Algorithm



Forget Loss

$$L_f(\theta) = -\frac{1}{|\Omega_{\mathcal{F}}|} \sum_{(i,t) \in \Omega_{\mathcal{F}}} (p_{i,t}(\theta))^{a \cdot s_i^{-b}} \ell_{i,t}(\theta) \leftarrow -$$

Popular Fact / Strong Forgetting
Rare Fact / Slight Forgetting

Epoch-Level Retain Dual Update

$$\lambda_{k+1} = \Pi_{[0, \lambda_{\max}]} (\lambda_k + \eta_{\lambda} (\delta_k - \varepsilon))$$
$$\alpha_{k+1} = \Pi_{[0, \alpha_0 + \lambda_{\max}]} (\alpha_0 + \lambda_{k+1})$$

auto-balancing

Output

Q: What is the founded by Welch's?

A: Steve Jobs

Q: What is the capital of Poland?

A: Lublin

